Experiment-9

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Course Code: -MLA0305

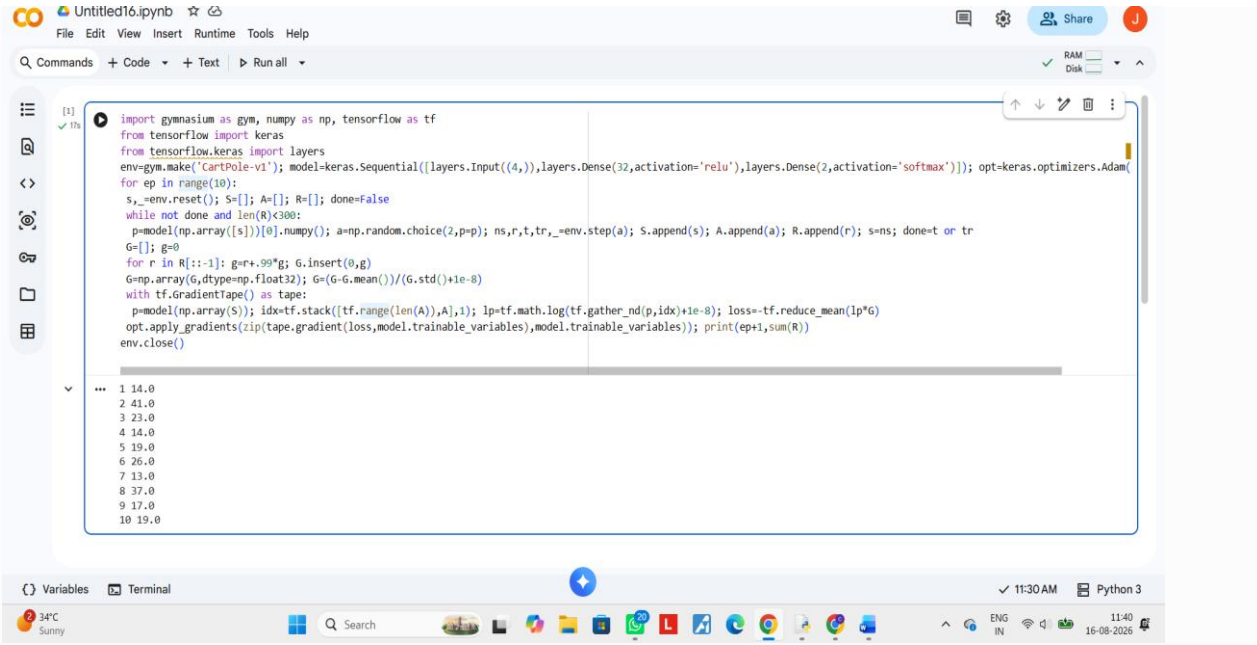
Title: Implementation of the REINFORCE Policy Gradient Algorithm

Aim:

To implement the REINFORCE algorithm and understand policy gradient methods for directly optimizing stochastic policies.

Procedure:

1. Select a suitable discrete-action environment.
2. Define a policy neural network.
3. Initialize policy parameters.
4. Generate complete episodes using the current policy.
5. Calculate the return for each time step.
6. Compute policy gradients using the REINFORCE objective.
7. Update the policy network using gradient ascent.
8. Repeat the process for multiple episodes.
9. Record episode rewards.
10. plot the learning curve.



Result:

The REINFORCE algorithm successfully learned a policy through direct policy optimization. The cumulative reward improved as training progressed.